

Kedar Panchal

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EDUCATION

Texas A&M University

Bachelor of Science in Computer Science

College Station, TX

August 2025 - May 2028

- **GPA:** 4.0
- **Relevant Coursework:** Software Engineering, Computer Systems, Cryptography, Data Structures and Algorithms, Computer Organization, Program Systems and Design, Linear Algebra, Statistics

WORK EXPERIENCE

Northrop Grumman

Software Engineer Intern, Mission Systems

San Antonio, TX

June 2026 - August 2026

- Updated legacy C code to eliminate memory vulnerabilities by architecting and implementing agentic code translation system.
- Reduced agent hallucinations by developing self-improving feedback loop that evaluated code and created test cases.
- Automated documentation generation, code linting, and deployment by designing GitLab CI/CD pipelines.
- Reduced deployment time by creating Ansible playbook to configure multi-container Podman environment on SELinux server.
- Communicated project progress to executive stakeholders and demonstrated product to senior leadership.

Texas A&M University

Research Assistant, Department of Computer Science

College Station, TX

January 2026 - Present

- Conducted research on planning algorithms for blind robots with limited sensing capabilities.
- Validated algorithm by implementing simulation environment in C++ and CGAL.

Firelight Innovations

AI Engineer

College Station, TX

September 2025 - January 2026

- Automated live website development during client calls by developing LangChain agent with video conferencing integration.
- Improved response time and context retrieval by connecting to LLM-optimized databases such as Redis and ChromaDB.

MathNMore LLC

Automation Developer

Austin, TX

July 2022 - July 2025

- Reduced loan payment notification time by 97.5% by developing JavaScript programs to automate late notice issuance.
- Tracked insurance payments for properties by writing JavaScript programs to interface with existing spreadsheet infrastructure.

PROJECTS

IMC Prosperity 4

April 2026 - May 2026

- Analyzed commodity market data to place within top 2.6% of competitors by building dashboard with Python and Matplotlib.
- Classified competing informed traders and detected outliers by implementing k-means clustering and isolation tree algorithms.
- Extracted meaningful patterns from noisy time-series data by preprocessing with Fourier transforms and Haar wavelet transforms.

Mr. Breathe

January 2026 - January 2026

- Detected and classified respiratory diseases by inventing intelligent spirometer.
- Interfaced spirometer with Swift iOS app for real-time data analysis using custom bluetooth code.
- Served data to app using Spring Boot backend and PostgreSQL database.

Hello World from Scratch

September 2025 - December 2025

- Developed bare-metal bootloader on ARM Cortex A53 chip to load and execute program from disk.
- Wrote FAT32 filesystem and VirtIO disk drivers in C.

GIMP AI Inpainter

January 2025 - March 2025

- Added generative AI inpainting capabilities to GIMP 3.0 by developing Python plugin that leveraged Hugging Face Diffusers.

SKILLS & CERTIFICATIONS

- **Programming Languages:** Bash, C/C++, HTML/CSS, Java, Python, SQL, TypeScript/JavaScript
- **Frameworks:** FastAPI, LangGraph, Next.js, React
- **Tools:** Docker, Git, GitLab CI/CD, GitHub Actions, Linux, PostgreSQL, Redis
- **Certifications:** Microsoft AI & ML Engineering, DeepLearning.AI Machine Learning Specialization, PCEP 30-02 Python Programmer